

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COS 202 — Past Questions & Materials

by Renance

Course: COS 202

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 6

Levels: 200

This study pack was compiled and restructured by Renance for COS 202 students at Federal University of Petroleum Resources, Effurun. It brings together 6 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 6 — Computer programming II Cos 202/221 2020/2021

Level 200 • Second Semester • PastQuestion

COURSE CODE: CSC221

DURATION: 2HRS SONINS

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Program that. displays @ temperature Conversion of °F to °C and absolute values chart an the

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Write a program that finds and prints the sum of all integers from 1 to n. mks

continues to read pairs of integers until it reads a pair with the

air is evenly divisible bythe second . 4mks

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, what is the delay between the time a radio signal leaves Earth Cog i

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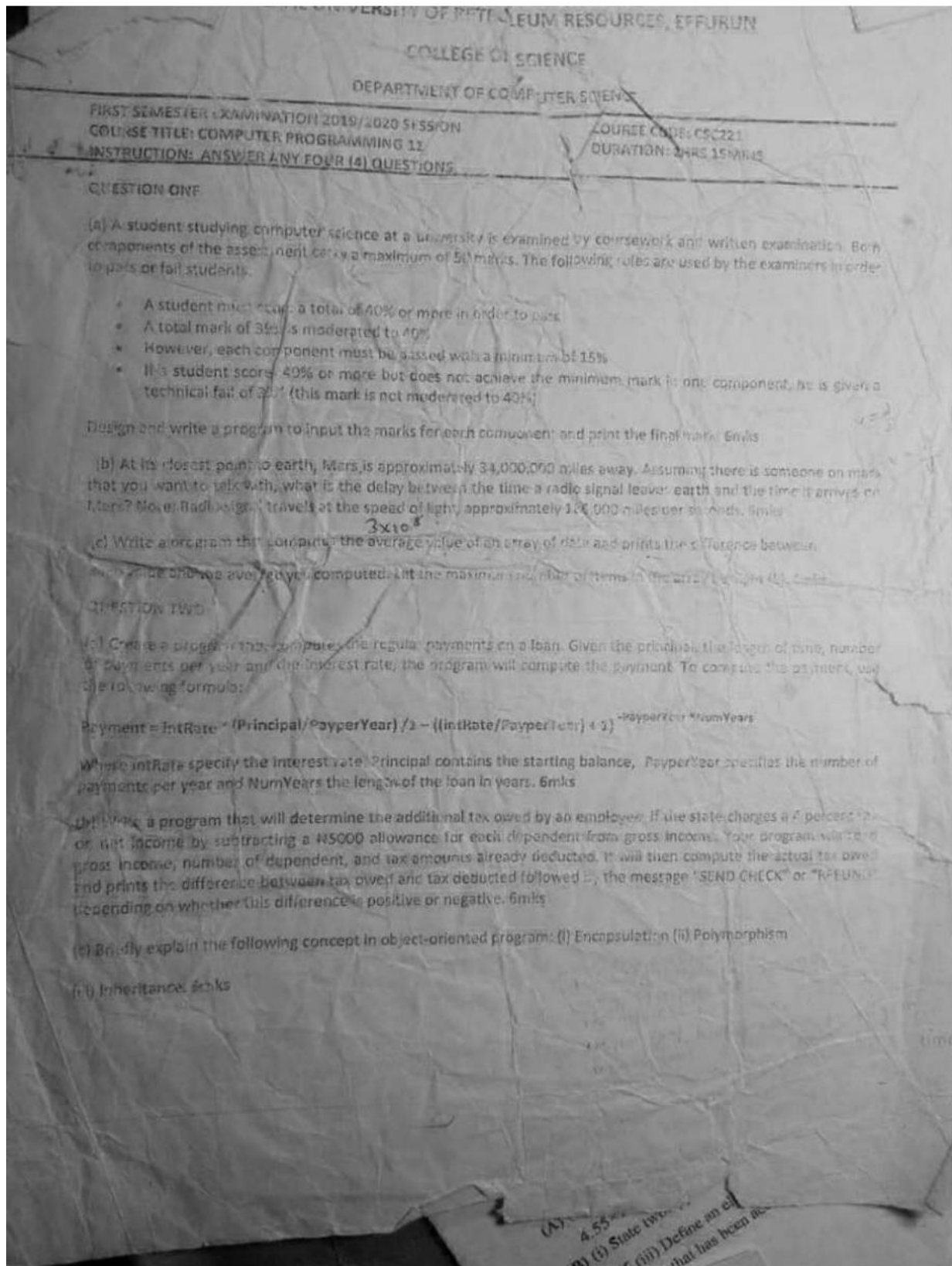
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Paper 2 of 6 — Computer programming II Cos 202/221 2019/2020 PastQuestion

Level 200 • Second Semester • PastQuestion

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QUESTION THREE

- A leap year is any year divisible by 4 unless it is divisible by 100, but not 400. Write a program to tell whether a year is a leap year. 6mks
- Write a C++ to find the length of the third side of a triangle, if we know the length of the other two sides (b and c) of a triangle and the angle between them in degrees (alpha). 6mks
- There are 9870 people in Ugbromu town, whose population increases by 10% each year, write a program using a **while** loop to determine how many years it will take for the population to exceed 30,000. 6mks

QUESTION FOUR

- Write a simple program in C++ that test the function that implements the Binary Search Algorithm. 6mks
- Write a C++ program that accumulates the sum of exams scores of up to 10 students, setting a sentinel of -1 and using a **while** statement. Let the program displays its results using formatted output. 6mks
- State the C++ rule for expression of evaluation. 6mks

QUESTION FIVE

- Your Physics Professor has asked you to write a program that displays the effect of gravity on the free-falling object. The program must display a table showing the height of an object dropped from a tower for every second that the object is falling. 6mks
- At its closest point, earth is approximately 34,000,000 miles away. Assuming there is someone on Mars that that you want to talk with, what is the delay between the time a radio signal leaves earth and the time it arrives on Mars.

Note: Radio signals travels at the speed of light, approximately 186,000 miles per seconds. Display the delay in terms of seconds and minutes. 6mks

- Using a **for** statement write a program that finds and prints the sum of all integers from 1 to n. 6mks

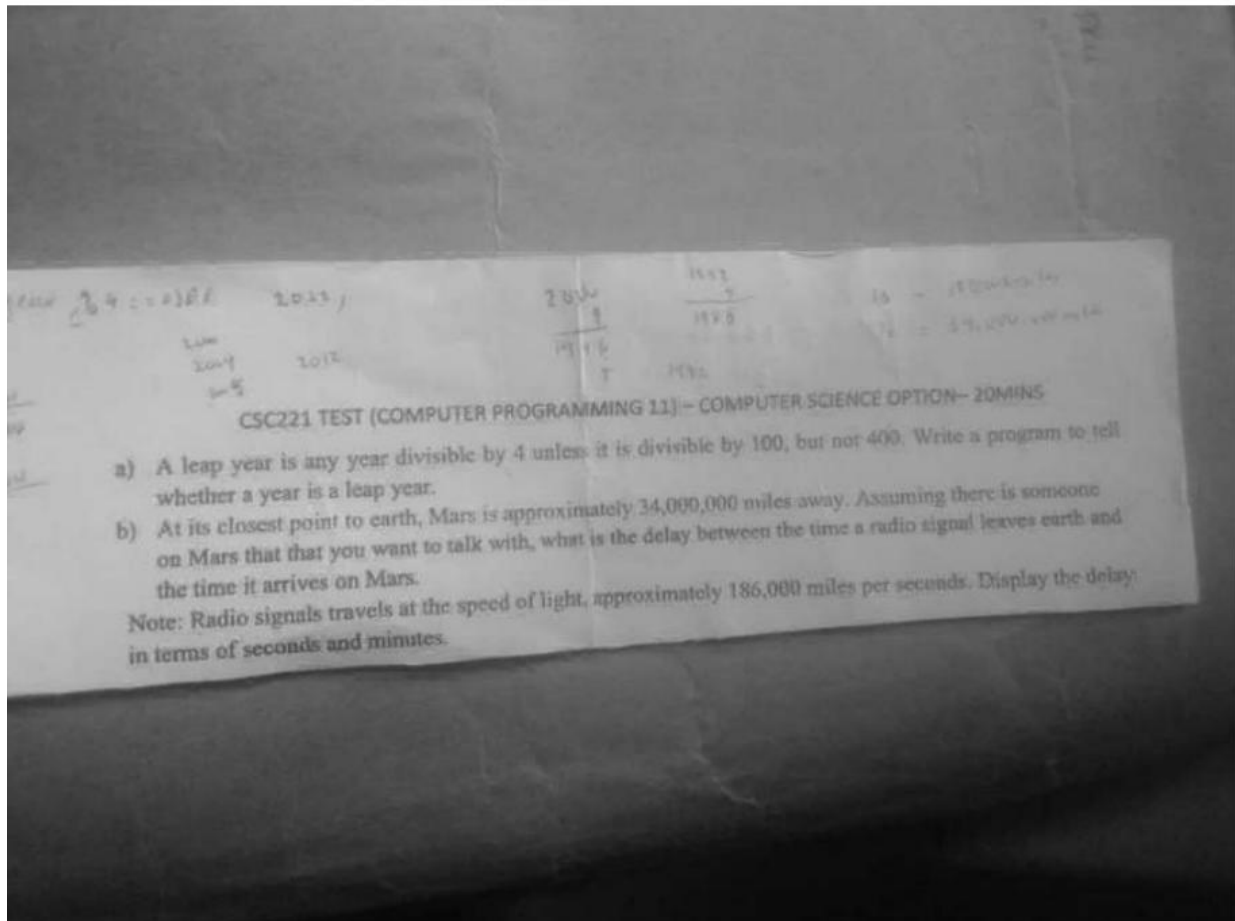
Handwritten notes and calculations:

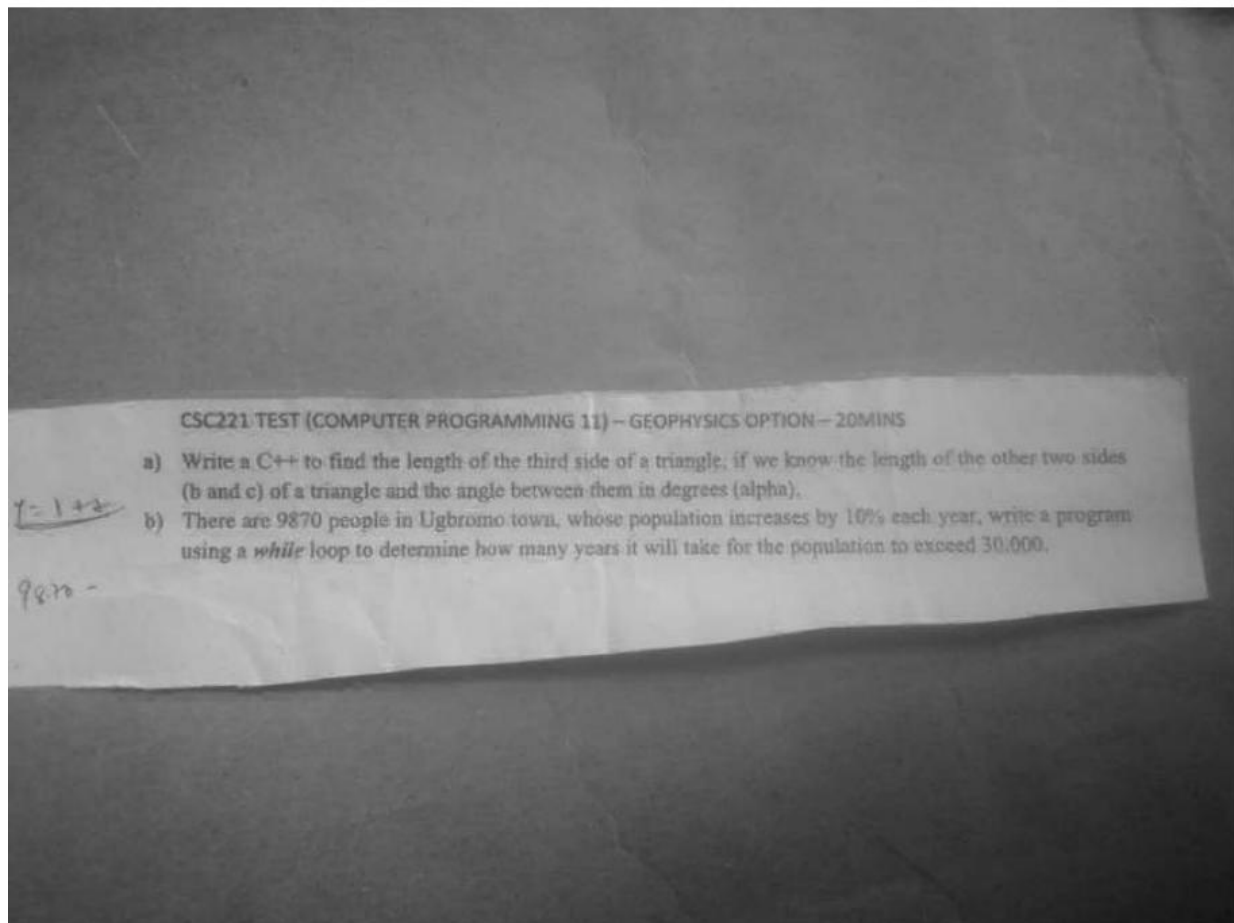
- 70 by 0.1
- 9870
- 9870 x 0.1 = 987
- Int population = 9870
- let percentage = 0.1
- for count++
- count
- 9870 x 0.1 = 987
- 9870 every by 0.1
- 9870 + n = 30,000
- Years = pop 987
- n = 20130
- 9870 x 10
- population = (population * percentage)
- Sum
- $g^h + 12$
- $g^h + 12$
- const float g = 9.8 ms⁻²
- while (t < 10)
- while (t < 10)
- for (i = 0; i < n; i++)
- sum
- sum

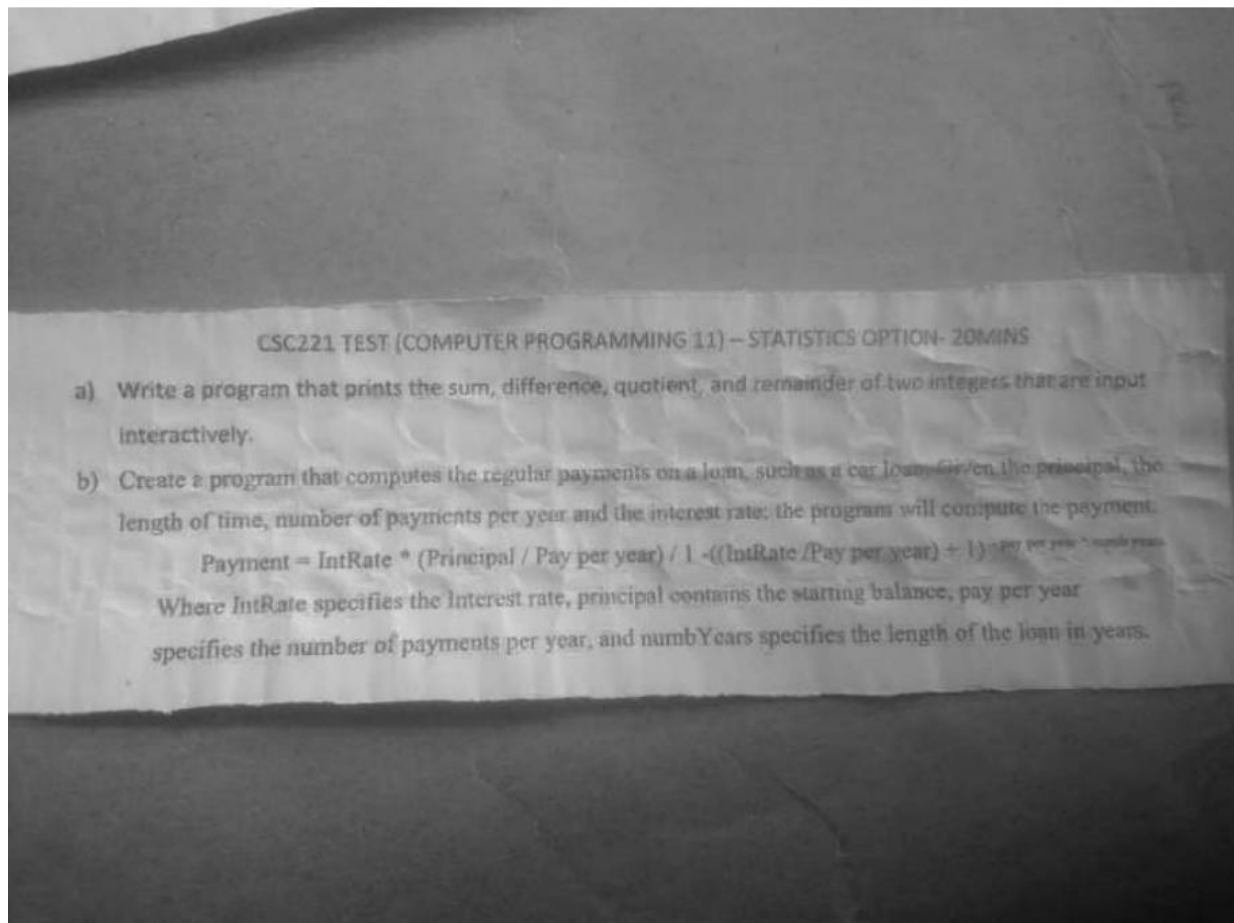
Paper 3 of 6 — Computer programming II Cos 202/221 Test Questions

Level 200 • Second Semester • PastQuestion

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CSC221 TEST: PHYSICS OPTION

(1) What does the following fragment print?

```
for(int i = 0; i < 10; i++) {  
    cout << i << " ";  
    if(i % 10 == 0) continue;  
    cout << "\n";  
}
```

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

(2) Write a simple program to test the divisibility of two integer numbers entered into the computer by the user. Use the if-else statement

CSC221: COMPUTER PROGRAMMING II TEST (PHYSICS)

- a) Write an *if* statement that might be used to compute the *average* of a set of n numbers whose sum is *total*, when n is greater than 0 and prints an error message when n is not greater than 0. The average should be computed by dividing by n
- b) Write a program segment that computes and displays $1+2+3+\dots+(n-1)+n$, where n is a data value

Paper 4 of 6 — Computer programming II Cos 202/221

PastQuestion 2023/2024

Level 200 • Second Semester • PastQuestion

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Population increases by 10% each year, write a

Program for the population to exceed 30,000. 6mks

by 100, but not 400. Write a program to check whether

0) miles away, Assuming there is someone on Mars

the time a radio signal leaves earth and the time it

Note: Radio signals travel at the speed of light

approximately 186,000 miles per second. Display the delay in

seconds.

4. Your company pays its hourly workers once a week.

Calculate the total pay for an employee who has worked (to the nearest half hour) and the employee's hourly pay rate, Weekly hours exceeding 40 are paid at a

rate of time and a half. Employees who earn over 2.00 per hour receive a bonus of 10% of their weekly earnings. Write a program to calculate the total pay for an employee. 6mks

5. Write a program that will determine the gross pay and net pay for an employee. 6mks

6. Write a C++ program that accumulates the total of exam scores of 10 students, setting a sentinel value to stop the loop.

7. Write a program that uses a while statement to calculate the sum of the first 10 natural numbers. 6mks

8. An employee's pay is based upon the number of hours

worked. Write a program that calculates the total pay for an employee. 6mks

9. Using a for statement write a program that finds and prints the sum of the first 10 natural numbers. 6mks

10. Write a program that calculates the sum of the first 10 natural numbers. 6mks

11. Create a program that computes the regular payments on a loan, such as a car loan. Given the principal, the

length of time; number of payments per year and the interest rate; the program will compute the payment,

$\text{Payment} = \frac{\text{InterestRate} * (\text{Principal} / \text{Payments per year})}{1 - (\text{InterestRate} / \text{Payments per year})} + \text{Principal}$

Where InterestRate specifies the Interest rate, principal contains the starting balance, r year specifies the

length of time in years, and Payments per year specifies the number of payments per year. 6mks

12. A cyclist coasting on a level road slows down from 10m/s to 2.5m/s in 10 seconds. Write a program that

calculates the cyclist's constant rate of deceleration and determine how long it will take the cyclist to come to

rest, given an initial speed of 10m/s. 6mks

13. Write a program that reads an unspecified number of numbers and prints them back with their deviation

from the mean. 6mks

14. Write a program that calculates the sum of the first 10 natural numbers. 6mks

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Write a program to read in a collection te eaptsngah Aoetil sores (OID) oat the naar OE
int the number of rege ould also display the category of each score read in. 6ii)ks 4
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4 Sum = 0;

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Paper 5 of 6 — COS 202 Computer Programming 2023/2024

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE (DEPARTMENT OF COMPUTER SCIENCE)

SECOND SEMESTER EXAMINATION 2023/2024 SESSION

COURSE CODE: CSC221

COURSE TITLE: COMPUTER PROGRAMMING 11

INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS

Time Allowed: 2HRS 15 MINS

QUESTION ONE

- (a) Write a program that will determine the additional state tax owed by an employee. If the State charges a 4% tax on net income by subtracting a \$5000 allowance for each dependent from gross income. Your program should read gross income, number of dependent, and tax amounts already deducted. It will then compute the actual tax owed and print the difference between tax owed and tax deducted followed by the message "SEND CHECK" or "REFUND" depending on whether this difference is positive or negative. 9mks
- (b) Write a simple C++ program that computes both the summation and factorial of the numbers 1 through 50. Declare the loop variable i inside the for loop. 4mks
- (c) Write a C++ to find the length of the third side of a triangle, if we know the length of the other two sides (b and c) of a triangle and the angle between them in degrees (alpha). 5mks

QUESTION TWO

- (a) There are 9870 people in Ugbromo town, whose population increases by 10% each year, write a program using a **while** loop to determine how many years it will take for the population to exceed 30,000. 6mks
- (b) A leap year is any year divisible by 4 unless it is divisible by 100, but not 400. Write a program to tell whether a year is a leap year. 6mks
- (c) At its closest point to earth, Mars is approximately 34,000,000 miles away. Assuming there is someone on Mars that you want to talk with, what is the delay between the time a radio signal leaves earth and the time it arrives on Mars.
- Note: Radio signals travels at the speed of light, approximately 186,000 miles per seconds. Display the delay in terms of seconds and minutes. 6mks

QUESTION THREE

- (a) Your company pays its hourly workers once a week. An employee's pay is based upon the number of hours worked (to the nearest half hour) and the employee's hourly pay rate. Weekly hours exceeding 40 are paid at a rate of time and a half. Employees who earn over \$100 a week, must pay union dues of \$15 per week. Write a payroll program that will determine the gross pay and net pay for an employee. 8mks
- (b) Write a C++ program that accumulates the sum of exams scores of up to 10 students, setting a sentinel of -1 and using a **while** statement. 5mks
- (c) Using a **for** statement write a program that finds and prints the sum of all integers from 1 to n. 5mks

QUESTION FOUR

- (a) Create a program that computes the regular payments on a loan, such as a car loan. Given the principal, the length of time, number of payments per year and the interest rate; the program will compute the payment.
- $\text{Payment} = \text{IntRate} * (\text{Principal} / \text{Pay per year}) / 1 - ((\text{IntRate} / \text{Pay per year}) + 1) - \text{pay per year} * \text{numb years}$
- Where *IntRate* specifies the Interest rate, *principal* contains the starting balance, *pay per year* specifies the number of payments per year, and *numbYears* specifies the length of the loan in years. 6mks
- (b) A cyclist coasting on a level road slows down from 10m/hr to 2.5m/hr in a minute. Write a program that calculates the cyclist constant rate of deceleration and determine how long it will take the cyclist to come to rest, given an initial speed of 10m/hrs. 6mks
- (c) Write a program that reads an unspecified number of numbers and prints them together with their deviation from the mean. 6mks

QUESTION FIVE

- (a) Write a program to read in a collection of exams scores ranging from 0 to 100. Your program should count and print the number of outstanding scores (90-100), number of satisfactory scores (60-89) and the number of unsatisfactory scores (0-59). It should also display the category of each score read in. 6mks
- (b) What is displayed by the segment.
- ```
Sum = 0; student = 1
while (sum <= 100)
 sum += 5;
 cout << sum << endl;
```
- Rewrite the loop so that it prints all multiples of 5 from 0 through 100 inclusive. 6mks
- (c) State the C++ rule for expression of evaluation. 6mks

Exam Score  $\geq 0 \text{ } 88 \leq 100$

for (score  $\geq 0 \text{ } 88 \leq 100$ )  
Enter Score

tax owed = Gross income - allowance  
allowance = 5

## Paper 6 of 6 — COS 202 PQ 2024

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